Arista Access Node KR-ACC-24 Routing Protocol Flap SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_8031

Description

This SOP addresses the root cause of a BGP route churn, OSPF adjacency loss, and ISIS LSP flooding failure on the Arista Access Node KR-ACC-24 at the KR Puram Access site. The routing protocol flap failure is attributed to the inherent nature of dynamic routing protocols, such as BGP, OSPF, and ISIS, which continuously exchange routing information between nodes to ensure optimal path selection. However, in the event of a topology change, network failure, or configuration anomaly, these protocols can exhibit instability, leading to excessive route flapping. This instability can be exacerbated by the Arista Access Node's role as a key component in the network's routing hierarchy, as it must maintain consistent routing information across multiple links and protocols. Furthermore, the node's vendor-specific configuration, including the use of Arista's EOS operating system, may introduce additional complexity and variability in the routing protocol behavior. To mitigate this issue, it is essential to employ a systematic approach to troubleshooting and resolving the root cause of the routing protocol flap failure.

Pre-requisite Checks

- Step: Verify Node Availability, Description: Ensure that the KR-ACC-24 node is reachable and operational before proceeding with the SOP.

- Step: Check Redundant Link Status, Description: Verify that the redundant links connected to the KR-ACC-24 node are functioning correctly and providing a stable routing path.

- Step: Dispatch Remote Hands, Description: Dispatch remote hands to the KR Puram Access site to assist with troubleshooting and hardware access if required.

Potential Root Causes

Cause	Probability
Misconfigured BGP Neighbor	High
OSPF Adjacency Loss due to Link Flapping	Medium
ISIS LSP Flooding due to Hardware Issues	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays a summary of the BGP routing information, including the number of established sessions, total routes, and any errors or warnings. This output helps identify any issues with BGP neighbor relationships or routing information.
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Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays information about the OSPF neighbor relationships, including the state of the adjacency and any errors or warnings. This output helps identify any issues with OSPF adjacency formation or maintenance.

Command: show isis database, Expected Output: ISIS LSP database information, Description: This command displays the ISIS LSP database, including the LSPs, sequence numbers, and any errors or warnings. This output helps identify any issues with ISIS LSP flooding or hardware-related problems.
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Resolution Steps

Step 1: Step: Verify BGP Neighbor Configuration, Command: show bgp neighbor KR-ACC-24, Description: Verify that the BGP neighbor configuration is correct and consistent across all nodes in the network. Check for any errors or warnings in the output, and ensure that the neighbor relationship is established and stable.

Step 2: Step: Re-Advertise Routes, Command: clear bgp * advertise, Description: Clear the BGP advertise cache and re-advertise the routes to ensure that the routing information is updated and consistent across the network.

Step 3: Step: Adjust OSPF Interface Parameters, Command: configure replace ospf interface KR-ACC-24, Description: Adjust the OSPF interface parameters to optimize the adjacency formation and maintenance. This may involve changing the hello timer, dead timer, or other interface-specific settings.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP summary information, Description: Verify that the BGP routing information is stable and consistent across the network. Check for any errors or warnings in the output, and ensure that the neighbor relationship is established and stable.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: Verify that the OSPF neighbor relationships are stable and consistent across the network. Check for any errors or warnings in the output, and ensure that the adjacency is established and stable.

Command: show isis database, Expected Output: ISIS LSP database information, Description: Verify that the ISIS LSP database is stable and consistent across the network. Check for any errors or warnings in the output, and ensure that the LSPs are correctly sequenced and propagated.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: configure rollback to KR-ACC-24, Description: Restore the original configuration to the KR-ACC-24 node to revert to the previous stable state.

- Step: Re-Apply Configuration Changes, Command: configure replace KR-ACC-24, Description: Re-apply the configuration changes that were made to resolve the issue, ensuring that the node is back to its original state.